YUVRAJ SINGH

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LinkedIn | **GitHub** | **LeetCode** | **HackerRank** | **Portfolio**

PROFESSIONAL SUMMARY

GATE CS-IT 2026 Qualified Computer Science Engineering student specialising in **Artificial Intelligence & Machine Learning**, with hands-on experience building **RAG pipelines, low-latency LLM systems, AI agents, and full-stack AI applications**. Skilled in **C, C++, Python, Java, LangChain, Hugging Face, and Machine Learning**, with research experience at the **Indian Institute of Science (IISc)** and a strong foundation in **Data Structures, OOPS, and AI/ML**.

PROFESSIONAL EXPERIENCE

Summer Research Intern · *Indian Institute of Science (IISc), Bengaluru* Jun 2026 – Aug 2026
Bengaluru, India

- Developed **DislocationTrace**, an integrated web-based framework for automated dislocation trace analysis and Burgers vector determination from TEM diffraction images using computer vision and crystallographic analysis.
- Built a modular computational pipeline integrating image processing, diffraction-vector indexing, dislocation line reconstruction, and the $g \cdot b = 0$ invisibility criterion for Burgers vector identification.
- Conducted applied AI/ML research under faculty supervision, engaging with state-of-the-art literature and contributing to experimental computer vision and materials-science analysis pipelines.

AI Research Intern · *Intel Unnati Industrial Training* Jan 2026 – May 2026
Remote

- Architected and built an **AI Education Tutor for Remote India**, designing an end-to-end LLM pipeline for low-bandwidth and connectivity-constrained environments using **LangChain, Hugging Face Transformers, and Python**.
- Reduced LLM token consumption by **40%** through context pruning and payload optimisation, lowering inference overhead for resource-constrained deployments.
- Improved end-to-end response latency by approximately **30%** through asynchronous retrieval, efficient context management, and batched tokenisation.
- Led a **4-member cross-functional team** through a 12-week Agile development cycle, delivering a functional **Streamlit prototype** on schedule.

AI Virtual Intern – Batch 3 · *Infosys Springboard Internship 6.0* Sep 2025 – Nov 2025
Remote

- Selected for a 10-week **AI internship programme**, gaining hands-on experience across end-to-end data science pipelines, enterprise AI tools, and production software development workflows.
- Achieved a **100% task-completion rate** across assessments, hands-on labs, and capstone submissions, demonstrating consistent execution throughout the programme.

AI Agent Training Intern · *IBM* Jul 2025 – Aug 2025
Remote

- Completed a 4-week intensive programme in **Agentic AI, multi-agent orchestration, and autonomous workflow automation**, earning formal IBM certification.
- Designed and documented **2 AI agent workflows** demonstrating autonomous task delegation, structured inter-agent communication, and practical agent-based system design.

Technical Training Intern · *India Space Labs* Jun 2025 – Jul 2025
Remote

- Completed a 6-week technical training programme exploring **space technology, advanced drone systems, and AI applications**, earning formal certification.
- Analysed **3+ domain-specific datasets** across space-tech and AI, contributing insights to **2 team-based simulations** modelling real-world technology deployment scenarios.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, C, C++, HTML, CSS, SQL
AI / ML / DL: LangChain, Hugging Face, PyTorch, TensorFlow, Keras, scikit-learn, Streamlit, OpenCV, FAISS
Web & Backend: Node.js, Express.js, Spring Boot, Bootstrap, REST APIs
Data & Visualisation: Pandas, NumPy, Matplotlib, Seaborn
Databases: MySQL, MongoDB, Oracle AI Vector Search
Cloud & DevOps: AWS, Oracle Cloud, Docker, Vercel, Git, GitHub, NPM, Postman, CI/CD, CUDA
Design Tools: Figma
Domains: Machine Learning, Deep Learning, NLP, Computer Vision, Generative AI, AI Agents, Prompt Engineering, RAG, LLMs, High-Performance Computing
Languages (Human): English (Professional), Hindi (Native), Bengali (Native)

PROJECTS

RAG-Enhanced LLM Legal Assistant for India 2026 – Present
| *LangChain* | *FAISS* | *Hugging Face* | *LLM APIs* | *Streamlit* | *Git*

- Developing an **end-to-end RAG-based legal AI assistant** to retrieve and synthesise information from Indian legal documents, including IPC, CrPC, RTI, BNS, BNSS, and the Constitution of India.
- Building a **semantic retrieval pipeline** using Hugging Face embeddings and FAISS vector search to retrieve contextually relevant legal information beyond traditional keyword-based search.
- Implementing **document chunking, query reformulation, multi-step prompting, and retrieved-context grounding** to improve retrieval quality and reduce unsupported LLM responses.
- Building an accessible interface using **LangChain, LLM APIs, and Streamlit**.

AI Education Tutor for Remote India 2026
| *Python* | *LangChain* | *Hugging Face* | *LLM APIs* | *Streamlit* | *Git*

- Engineered a **low-latency LLM inference pipeline** for low-bandwidth environments, reducing token overhead by **∼40%** through context filtering and payload compression while improving response latency by **∼30%**.
- Built a **connectivity-resilient AI tutoring system** targeting Tier-3 and remote communities, incorporating efficient context management and graceful degradation for intermittent network conditions.

DislocationTrace: Automated TEM Dislocation Analysis 2026
| *Python* | *Computer Vision* | *Image Processing* | *Scientific Computing* | *Web Development* | *Git*

- Developed **DislocationTrace**, an end-to-end web-based framework that automates dislocation trace analysis and Burgers vector determination from **TEM diffraction images** using computer vision and crystallographic reasoning.
- Engineered a **modular computational pipeline** integrating image processing, diffraction-vector indexing, dislocation line reconstruction, and automated validation using the $g \cdot b = 0$ invisibility criterion.
- Applied **computer vision and algorithmic optimisation** to transform experimental microscopy data into structured analytical outputs, contributing to research workflows in materials science under faculty supervision at **IISc**.

EDUCATION

MCKV Institute of Engineering, West Bengal 2023 – 2027
B.Tech in Computer Science & Engineering — AI & ML **CGPA: 9.62 / 10**

Delhi Public School, Bokaro Steel City, Jharkhand
Class XII — PCM (CBSE) 70%

Saraswati Shishu Vidya Mandir, Bokaro Steel City, Jharkhand
Class X (Jharkhand Academic Council) 92%

Relevant Coursework: Data Structures & Algorithms, Operating Systems, OOP (Java), Data Analysis & Visualisation, Artificial Intelligence & ML (Deep Learning, NLP, Computer Vision, Neural Networks), Data Science & Analytics, Cloud Computing (AWS), High-Performance Computing (CUDA), DBMS (SQL), Networking, Data Security & Authentication, Data Mining

CERTIFICATIONS

- Oracle Cloud Infrastructure 2025 Generative AI Professional
- Oracle AI Vector Search 2025 Professional
- Oracle Cloud Infrastructure 2025 Data Science Professional
- Oracle Cloud Infrastructure 2025 AI Foundations Associate
- Google Cybersecurity Professional Certificate
- Introduction to AI & ML with Python — IBM

ACHIEVEMENTS & LEADERSHIP

- **GATE CS-IT 2026 Qualified** — Qualified one of India's most competitive postgraduate examinations in Computer Science & Information Technology.
- **Summer Research Fellowship Program (SRFP) 2026** — Selected for the prestigious research fellowship jointly conducted by the **Indian Academy of Sciences (IASc), Indian National Science Academy (INSA), and National Academy of Sciences, India (NASI)**; conducted research at the **Indian Institute of Science (IISc), Bengaluru** and participated in multiple scientific workshops at the **Raman Research Institute (RRI)**.
- **HackerRank Orchestrate Hackathon 2026** — Ranked **412 nationally**; built a multi-domain support triage agent in Python using **BM25 retrieval, domain classification, and LLM-based workflows**.
- **Leadership & Communities** — Community Member, Google Developer Group (GDG), MCKVIE Campus; Member, IETE; Campus Ambassador, E-Cell IIT Bombay; Active Member, Google Cloud & NVIDIA Community.